**Legend - Hierarchical machine learning identifies genomic features underlying serotype variation in *Vibrio cholerae***

**Graphical abstract:** Hierarchical machine learning-based serotype classification for Vibrio cholerae approach to investigate the overall population structure

**Alt text:** Graphical abstract showing the study methodology from the data retrieval, quality control, hierarchical machine learning and further downstream analysis.

**Figure:**

**Figure 1: Population structure analysis of 1,866 Vibrio cholerae genomes (A)** Distribution of VSP-I genomic islands among the PCoA clusters **(B)** Distribution of VSP-II genomic islands among the PCoA clusters **(C)** Distribution of serotypes among the PCoA clusters **(D)** Core genome phylogenetic tree with adjacent bars indicating the serotypes, PCoA clusters, VSP-I and II genomic islands, respectively

**Alt text**: Panel A shows complete VSP-I genomic islands predominantly in PCoA cluster A. Panel B shows intact and deletion variants of VSP-II, while panel C shows the predominance of O1 serotypes, Ogawa and Inaba, in cluster A. Panel D shows the core-genome phylogeny and associated annotations for serotype, PCoA cluster, and VSP-I and VSP-II status. The two predominant non-O1 clusters show residual, truncated, or absent genomic islands.

**Figure 2: Features for hierarchical machine learning-based classification. (A)** and **(C)** show the distribution of ROC-AUC values obtained by the logistic regression and random forest classifiers using feature sets selected by Boruta, LASSO, and their intersection for serogroup and serotype classification, respectively. **(B)** and **(D)** show the SHAP feature importance plots of the best-performing model for serogroup and serotype classification, respectively. Genes located within the OAGC regions are highlighted in blue.

**Alt text**: Panels A and C show consistently high ROC-AUC scores for serogroup classification and greater variability for serotype classification, respectively. Panels B and D show SHAP feature importance, highlighting OAGC genes in serogroup classification and wbeT sequence features, integrons, AMR and virulence genes, and plasmids in serotype classification.

**Figure 3: Heatmaps depicting the distribution of significantly associated (A) AMR, (B) virulence genes and (C) integrons among serotypes grouped based on PCoA clustering**

**Alt text:** Heatmaps show AMR genes in panel A, virulence genes in panel B, and integrons in panel C. PCoA cluster A shows enrichment of AMR genes, class 1 integrons, and toxin- and colonization-associated virulence genes.

**Table:**

**Table 1: Model performance for serogroup classification before and after 10 fold CV and hyperparameter tuning**

**Table 2: Model performance for serotype classification before and after 10 fold CV and hyperparameter tuning**

**Supplementary material:**

**Figures:**

Supplementary Figure 1 (Supplementary file-1): Mapping of misclassified genomes (false positives and false negatives) for (A) serogroup and (B) serotype classification.

**Alt text:** PCoA plots showing the distribution of misclassified genomes. Misclassified genomes in serogroup classification were located in cluster B, whereas those in serotype classification were distributed across both clusters A and B.

**Tables:**

Supplementary table 1 (Supplementary file-1): VSP genomic island classification criteria

Supplementary table 2 (Supplementary file 2):

Sheet 1: Chisq test association between AMR genes and serotype

Sheet 2: Chisq test association between virulence genes and serotype

Sheet 3: Chisq test association between integrons and serotype

Sheet 4: Chisq test association between transposons and serotype